Fewer Babies or Later Babies?

Lecture 1.7 – 2/10/26

Fertility Rate in U.S. Hit a Record Low in 2018

The rate of births fell again last year, according to new government data, extending a lengthy decline as women wait until they are older to have children.

By Sabrina Tavernise

Nov. 27, 2019, 12:00 a.m. ET

WASHINGTON — The fertility rate in the United States fell in 2018 for the fourth straight year, extending a steep decline in births that began in 2008 with the Great Recession, the federal government said on Wednesday.

There were 59.1 births for every 1,000 women of childbearing age in the country last year, a record low, according to the National Center for Health Statistics. The rate was down by 2 percent from the previous year, and has fallen by about 15 percent since 2007.

In all, there were 3,791,712 births in the country last year, the center said in its release of final birth data for 2018.

Fertility rates are essential measures of a society's demographic balance. If they are very high, resources like housing and education can be strai
postwar Baby Boom years. If they are too low, a c
people to replace its work force and support its el

In the United States, declines in fertility have not
immigration has helped offset them.

The country has been living through one of the lo
Demographers are trying to determine whether i
driven by deeper social change.

U.S. Fertility Rate Fell to a Record Low, for a Second Straight Year

The Mother Bomb

JAN. 14, 1979

Americans aren't making enough babies to replace ourselves

U.S. Fertility Rate Reaches a Record Low

Opinions

The demographic time bomb that could hit America

Roadmap

1. Some Definitions
2. Measures of fertility
   - CFR
   - GFR
   - ASFR
   - TFR
   - CCFR
3. Fertility, fecundity, and age

1. Definitions

- Fertility = number of children
- Infertility = reduced ability to conceive and bear a live child
  - *Medical definition:* “failure to achieve a pregnancy after 12 months or more of regular unprotected sexual intercourse”
- Fecundity = biological capacity to produce a live birth

Natural Fertility vs. Maximum Fertility

- Natural Fertility = the level of fertility in a population if fertility behavior is not altered according to the family size reached
- Maximum Fertility = the level of fertility in a population if fertility behavior is maximized

Maximum fertility??

Hutterites

- Emigrated to the US from Russia in 1870's
- Relatively closed to outside (no migration)
- Population grew from 443 to 8,542 from 1880-1990
  - (~19x; compare to 5x for USA)
- Considered birth control, including withdrawal, a sin
- Their fertility was seen as the "maximum"

Source: Ingoldsby and Stanton 1988

Maximum fertility??

TABLE 3. Selected Demographic Characteristics of Hutterite Women, by Age at Last Confinement (204 Couples)

	Age at last confinement (yr.)			
	Under 35	35–39	40–44	45 and over
Number of women	18	47	117	22
Mean age at first confinement (yr.)	22.0	22.5	22.1	22.2
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Source: Tietze 1957

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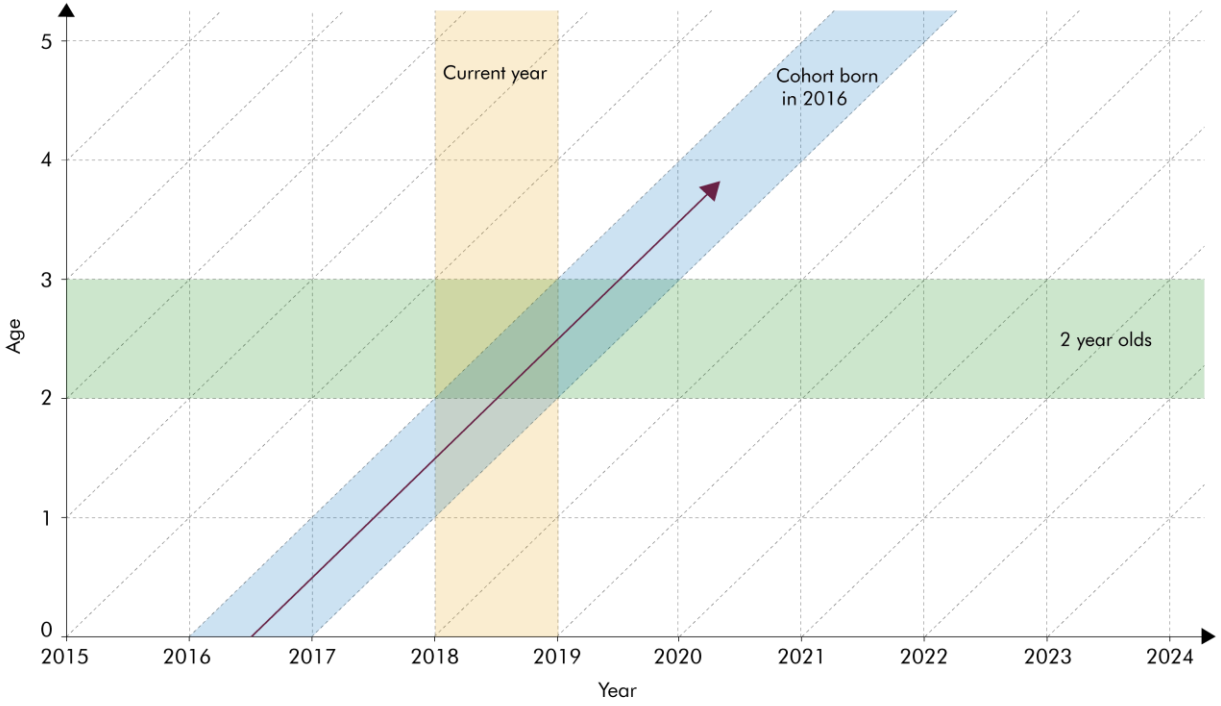
2. Measures of Fertility

1. Crude Fertility Rate
2. General Fertility Rate
3. Age-Specific Fertility Rate
4. Total Fertility Rate
5. Cohort Fertility Rate

Recall: 3 Concepts

1. **Age**
2. **Period**: Time (often year)
3. **Cohort**: Group born at the same time



	Pros	**Cons**
Period Measures		
Cohort Measures		

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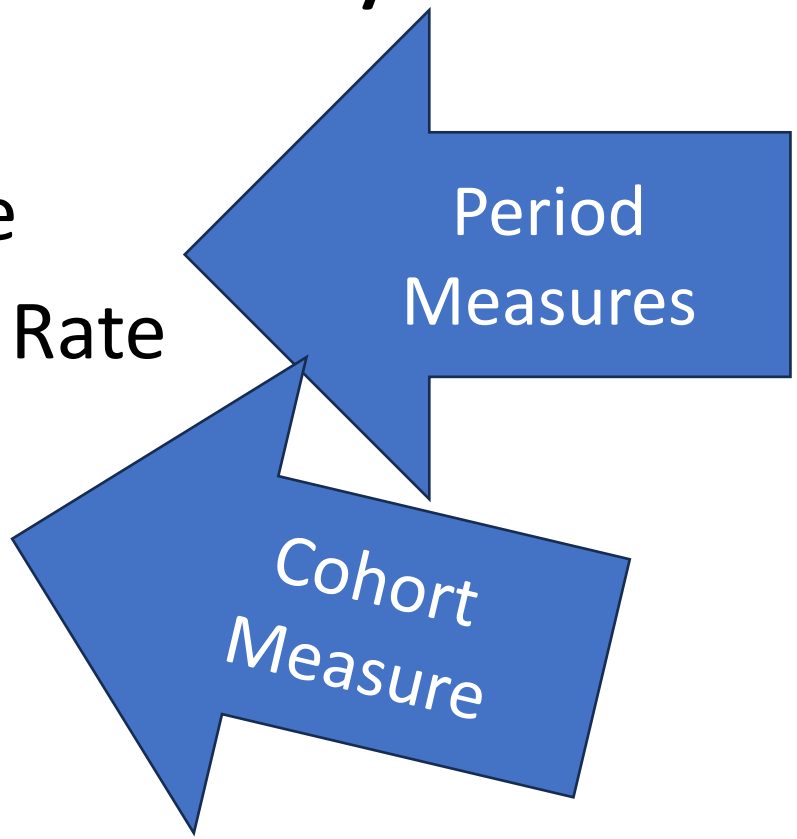

Period measures of fertility

- **Crude** Fertility Rate = # of births/# of people
- **General** Fertility Rate (GFR) = # of births/# of women of reproductive age
- **Age-Specific** Fertility Rate (ASFR) = # of births/# of women of a specific age
  - Usually a 5-year range
- **Total** Fertility Rate (TFR)

These are usually expressed per 1000 population (so x1000)

Example: Pennsylvania 1996

A. Number of live births: 147,890
B. Number of people: 12,220,500
C. Number of women of reproductive age: 2,583,748
D. Number of births to women aged 20-24: 30,613
E. Number of women aged 20-24: 403,191

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Crude Fertility Rate = A/B = (147,890/12,220,500)*1000 = 12.1

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GFR = A/C = (147,890/ 2,583,748)*1000 = 57.2

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$ASFR_{20\text{-}24}$ = D/E = (30,613/403,191)*1000 = 75.9

Total Fertility Rate (TFR)

- The total fertility rate (TFR) = the sum of the age-specific birth rates (ASFRs)
- Often estimated using *5*-year age groups.
  - Then the ASFRs are multiplied by *5*
- Can be interpreted as: the # of children a woman would expect to have if she lived through the *period's* age-specific fertility rates
- TFR is a *period* measure because it looks at fertility at a specific time period

Calculating TFR

AGE GROUP	NUMBER OF LIVE BIRTHS	POPULATION	ASFR
15-19	15,267	408,422	.0374
20-24	30,613	403,191	.0759
25-29	42,245	397,398	.1063
30-34	39,105	446,488	.0876
35-39	17,378	469,810	.0370
40-44	2,808	458,439	.0061
45-49	94	413,377	.0002

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		TFR = TOTALx5 =	1.7525

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		=	1.7525

Interpretation: If a woman were to live through the fertility rates of Pennsylvania in 1996, she would have 1.75 kids.

Problems with TFR

- “Tempo” vs. “Quantum”
  - Tempo = the timing of births
  - Quantum = the number of births

Problems with TFR

- “Tempo” vs. “Quantum”
  - Tempo = the timing of births
  - Quantum = the number of births
- Because TFR uses births at all ages at one time, cohorts *delaying* births *looks like* low fertility!
- This is called the “postponement transition”

The Postponement Transition

Two Demographic Mechanisms:

(1) Decline in Number of Kids *(Quantum)*

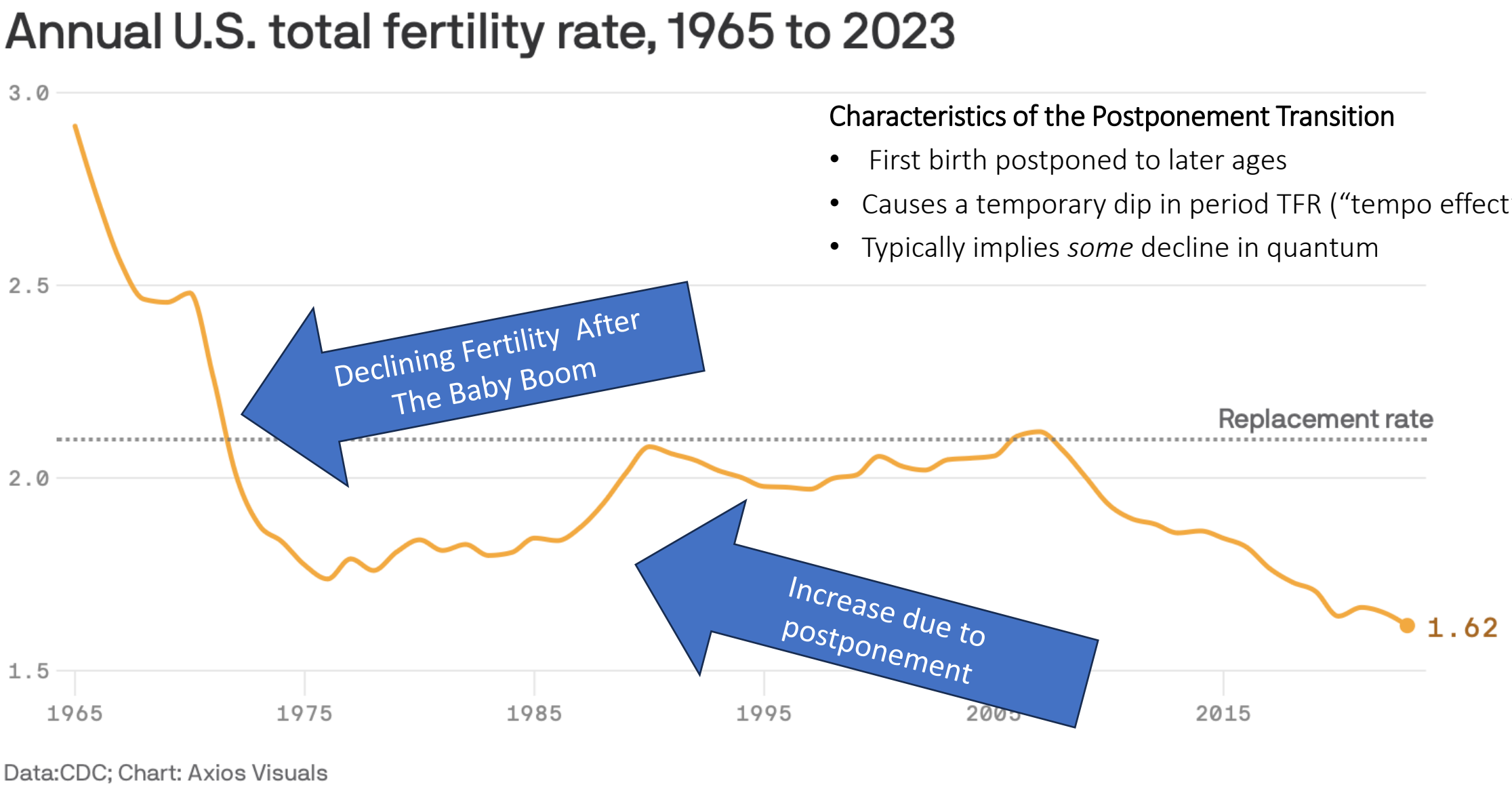
\+ (2) Shift in Timing of Kids *(Tempo)*

(3) Postponement Transition

Characteristics of the Postponement Transition

- First birth postponed to later ages
- Causes a temporary dip in period TFR ("tempo effect")
- Typically implies *some* decline in quantum

The Postponement Transition in the US



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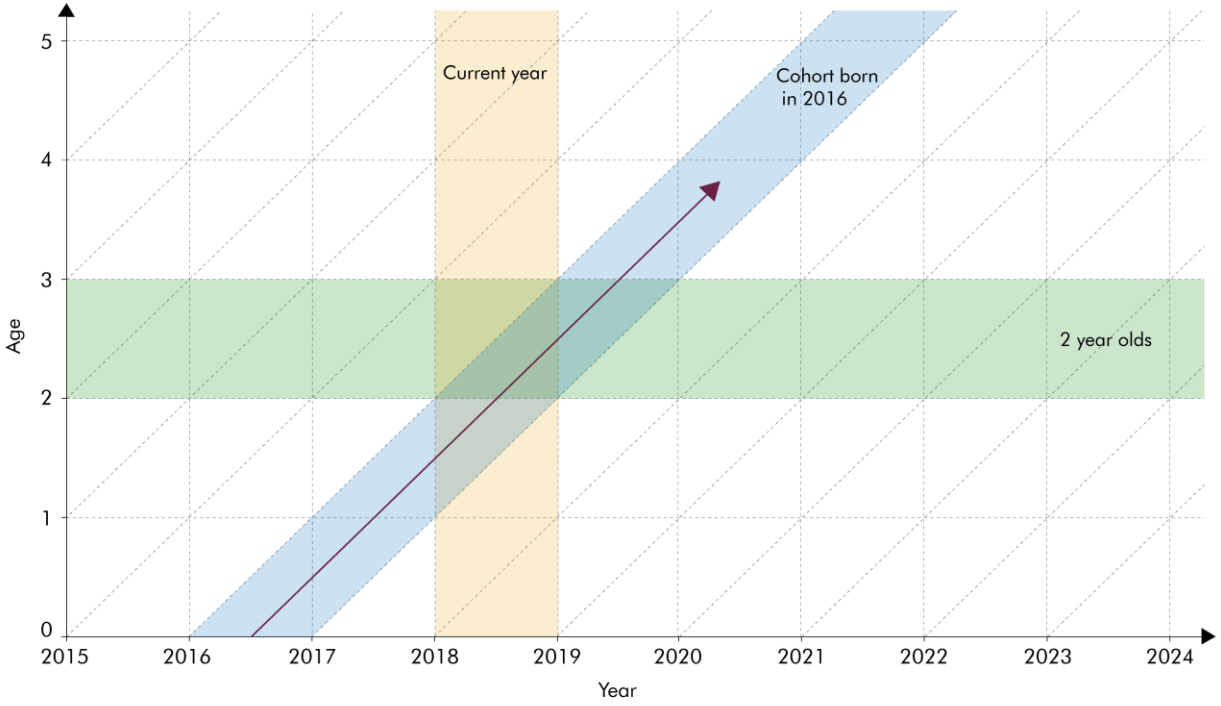


	Pros	**Cons**
Period Measures	Timely	May not be accurate
Cohort Measures		

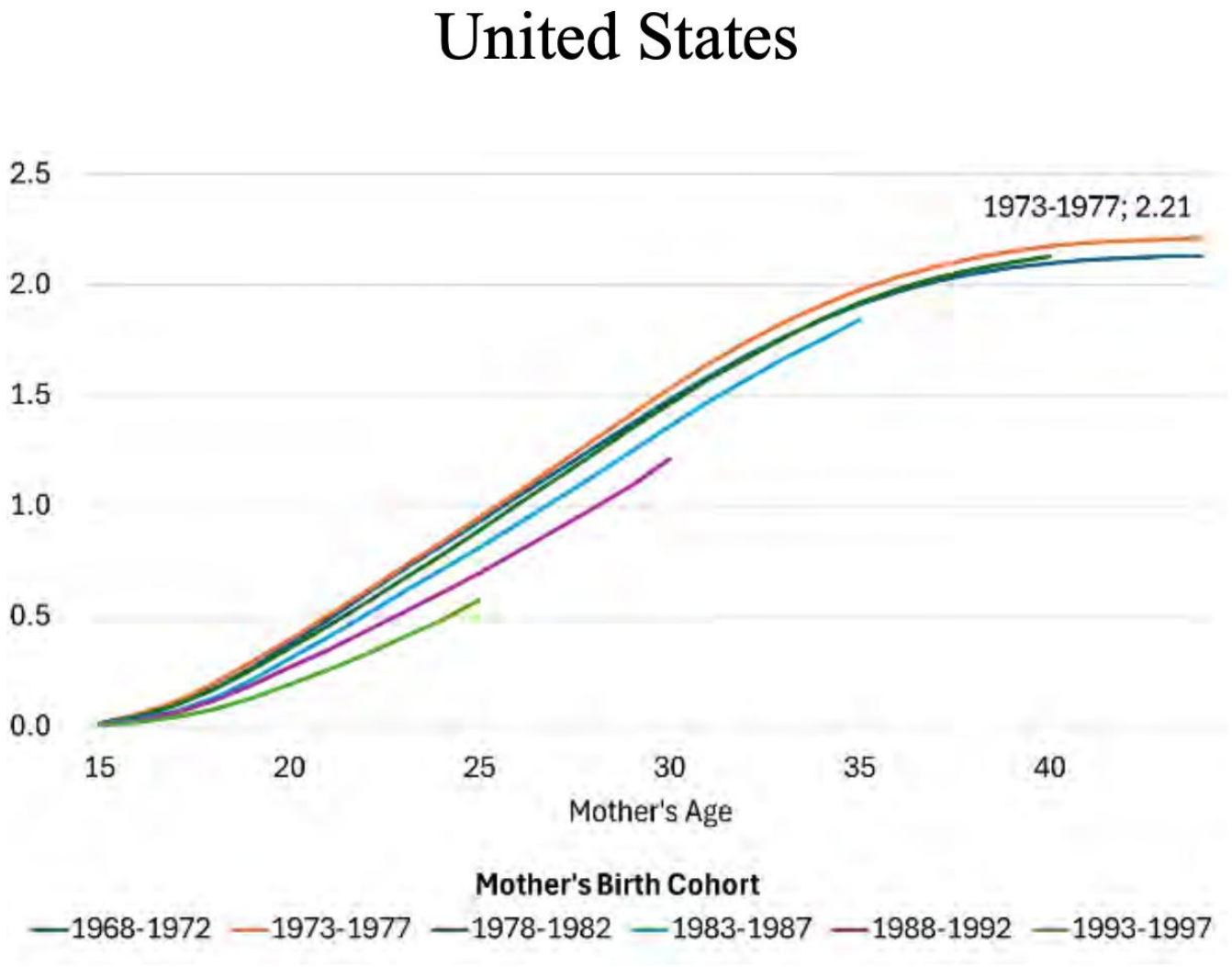
Cohort Fertility

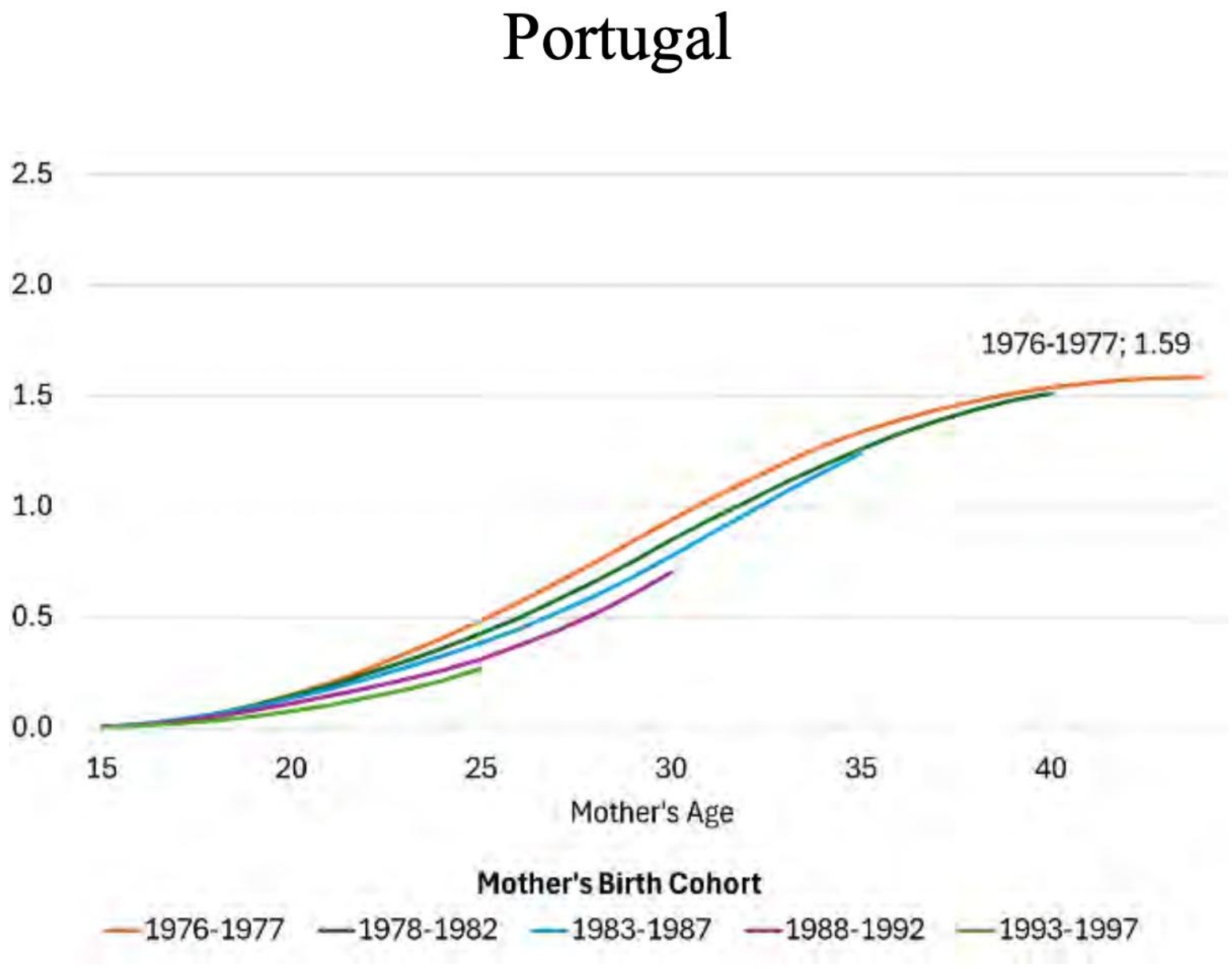
- Cohort Fertility Rate (CFR) = # of births/# of women in a cohort by a given age
- More “real” measure of fertility! ➕
- But we can’t know what will happen in the future! ➖

(These + and – are generally true of period vs. cohort measures!)

Cumulative Cohort Fertility:

Source: Kearney and Levine 2025





Very possible that younger cohorts will catch up!

(but we won't know until it happens!)

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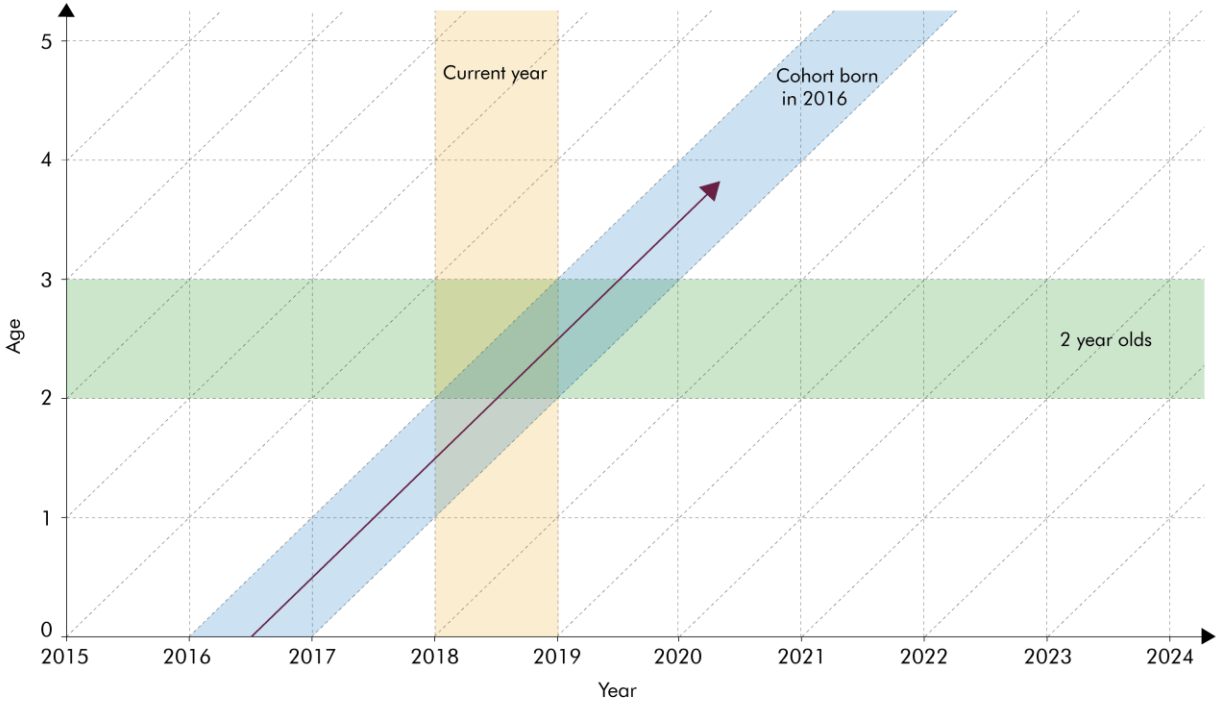
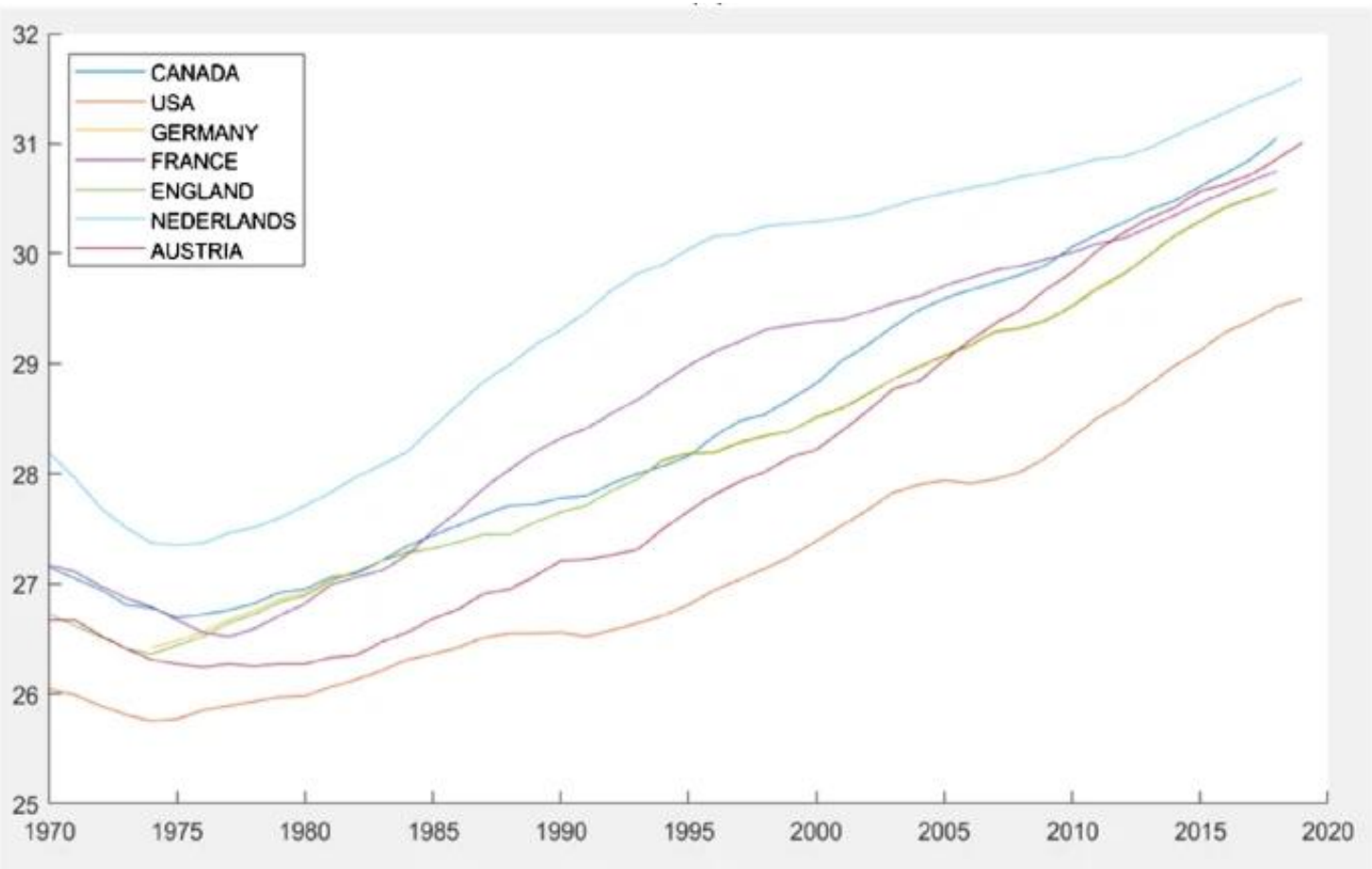


	Pros	**Cons**
Period Measures	Timely	May not be accurate
Cohort Measures	Accurate	Not timely (retrospective)

Fertility delay vs. decline around the world

Mean age at first birth

Dioikitopoulos & Varvarigos (2023)



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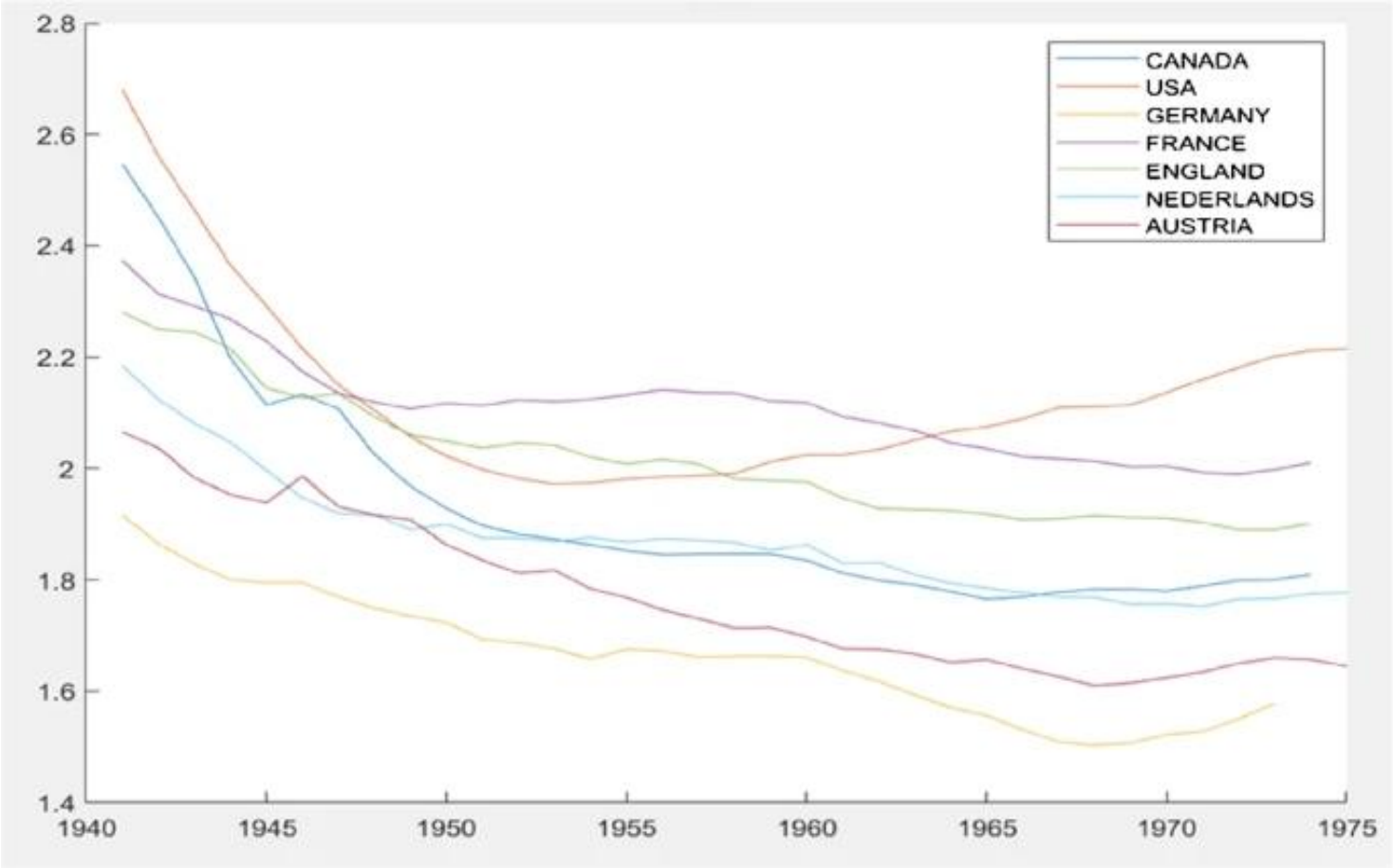
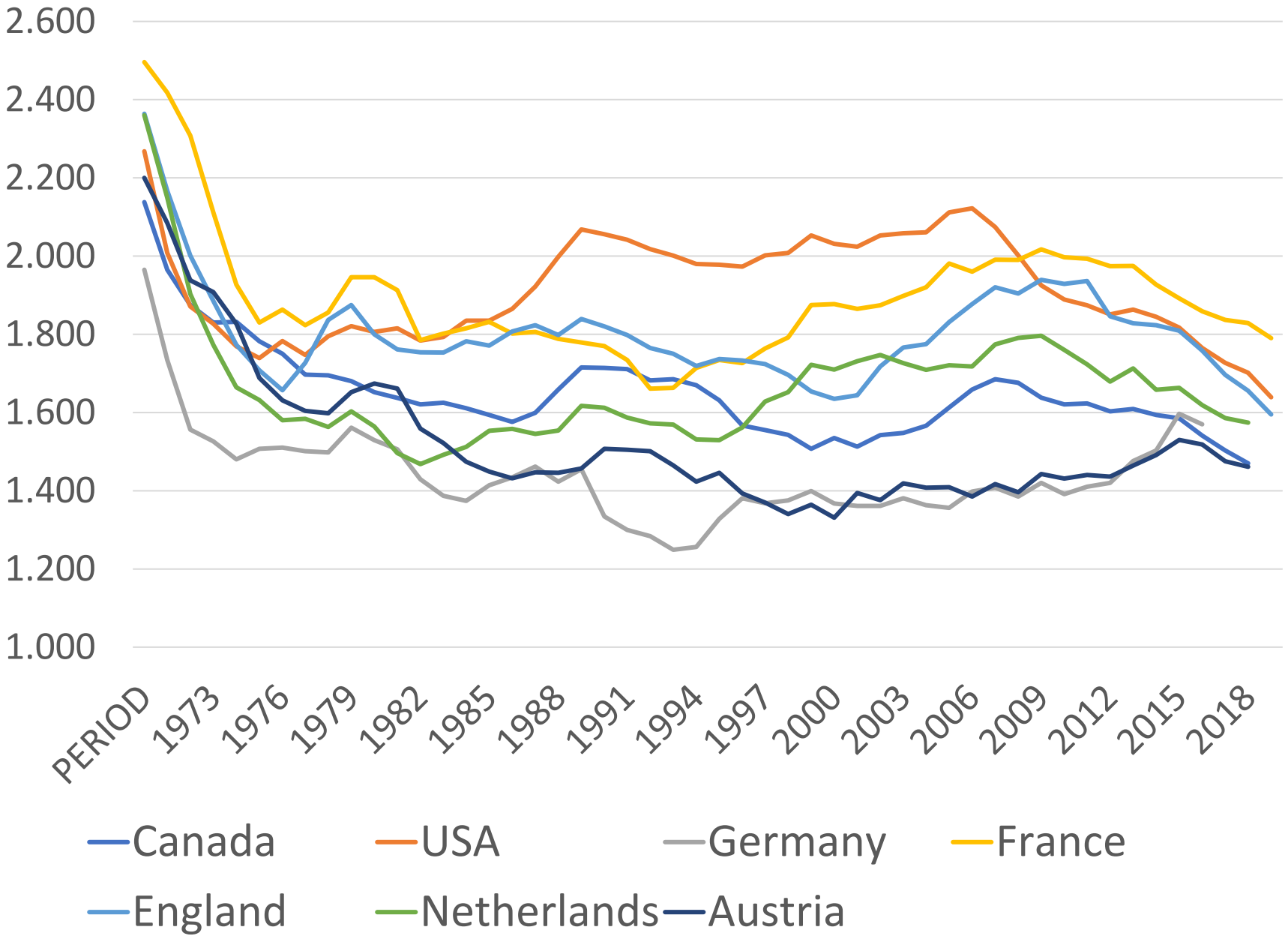


Cohort Completed Fertility

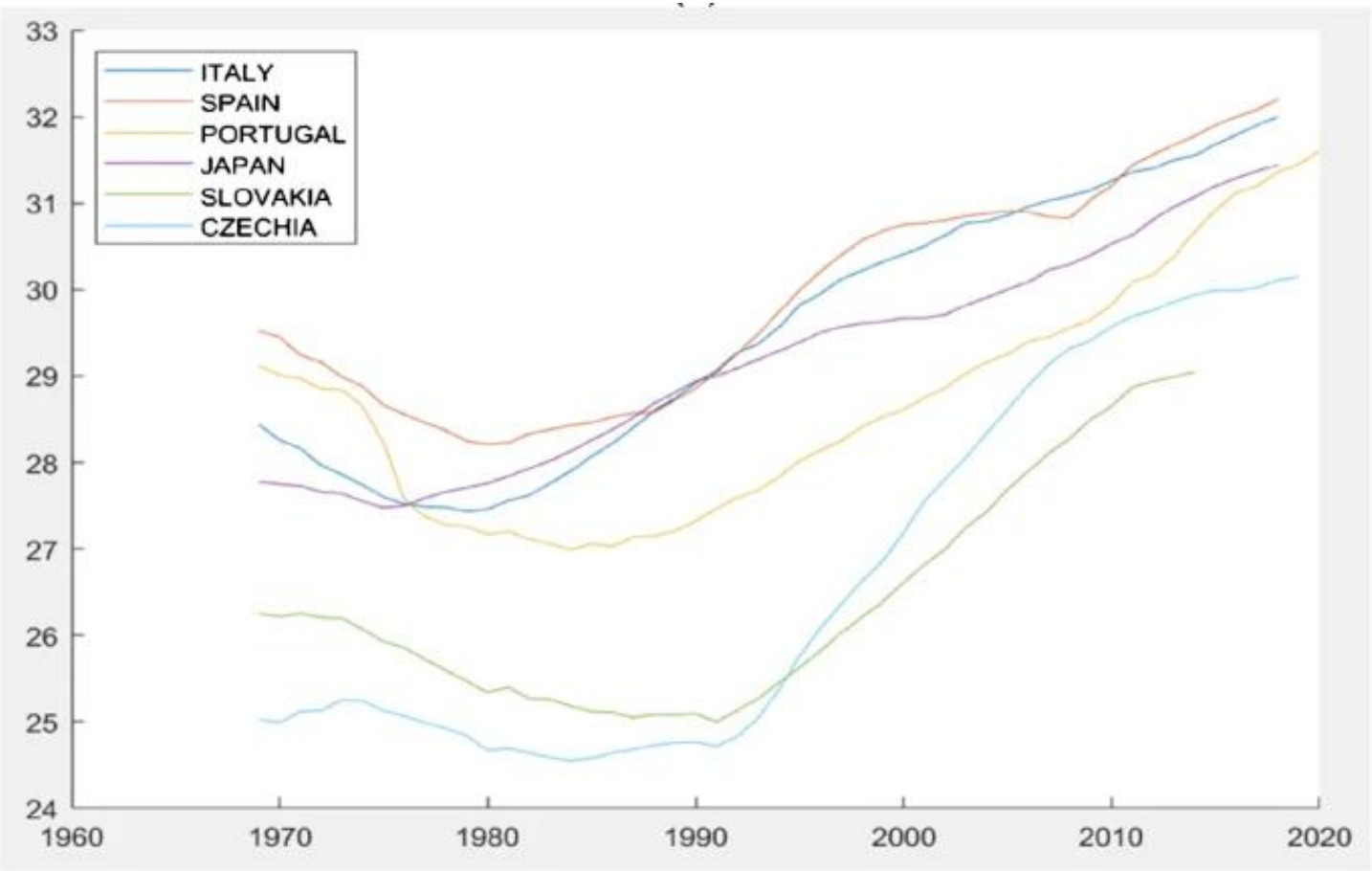
Dioikitopoulos & Varvarigos (2023)

Sometimes fertility delay leads to falling and then increasing TFR (postponement transition)

Human Fertility Database 2023

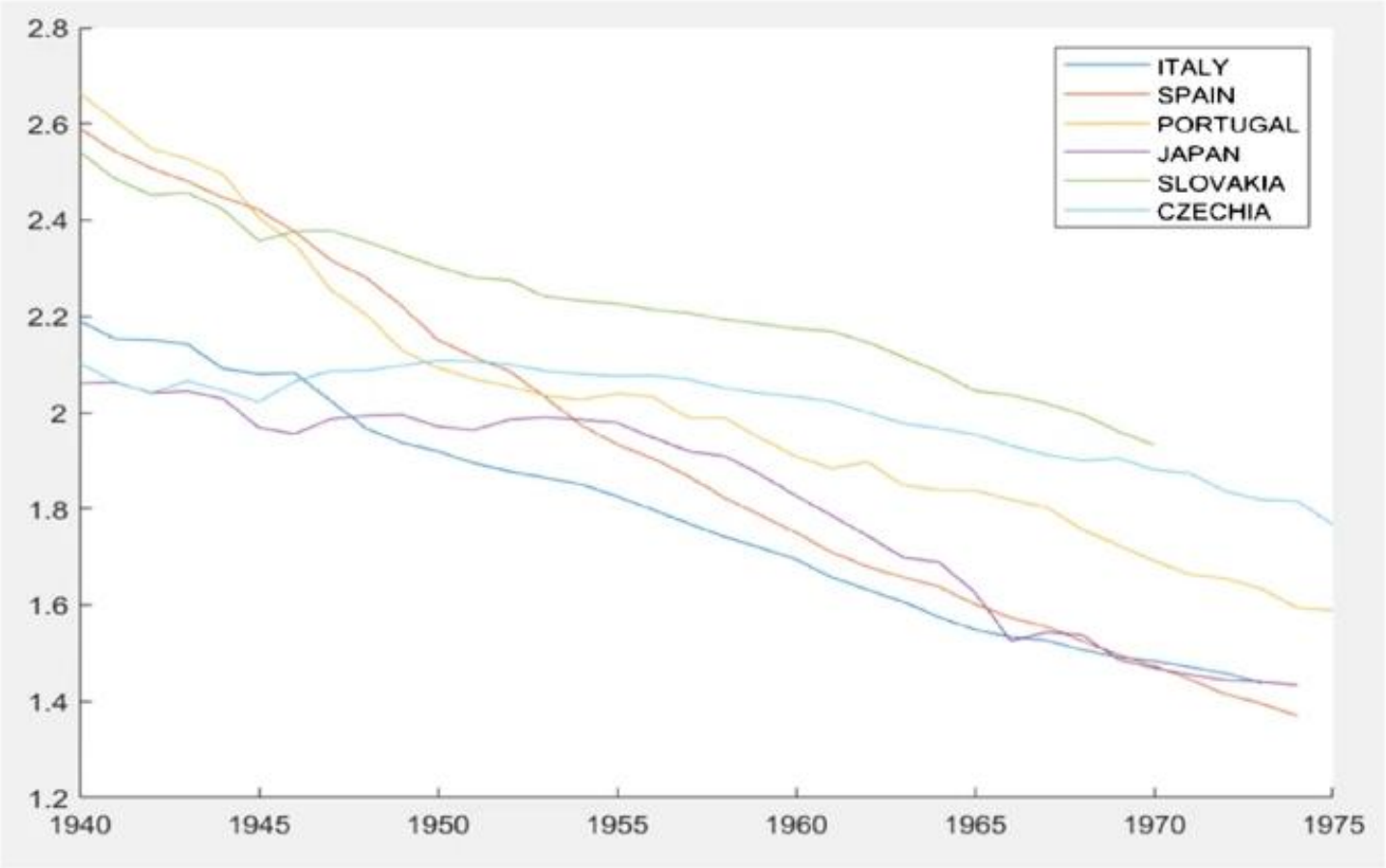


But in other cases age at first birth keeps going up…



Dioikitopoulos & Varvarigos (2023)

And completed cohort fertility keeps falling!



Cohort Completed Fertility

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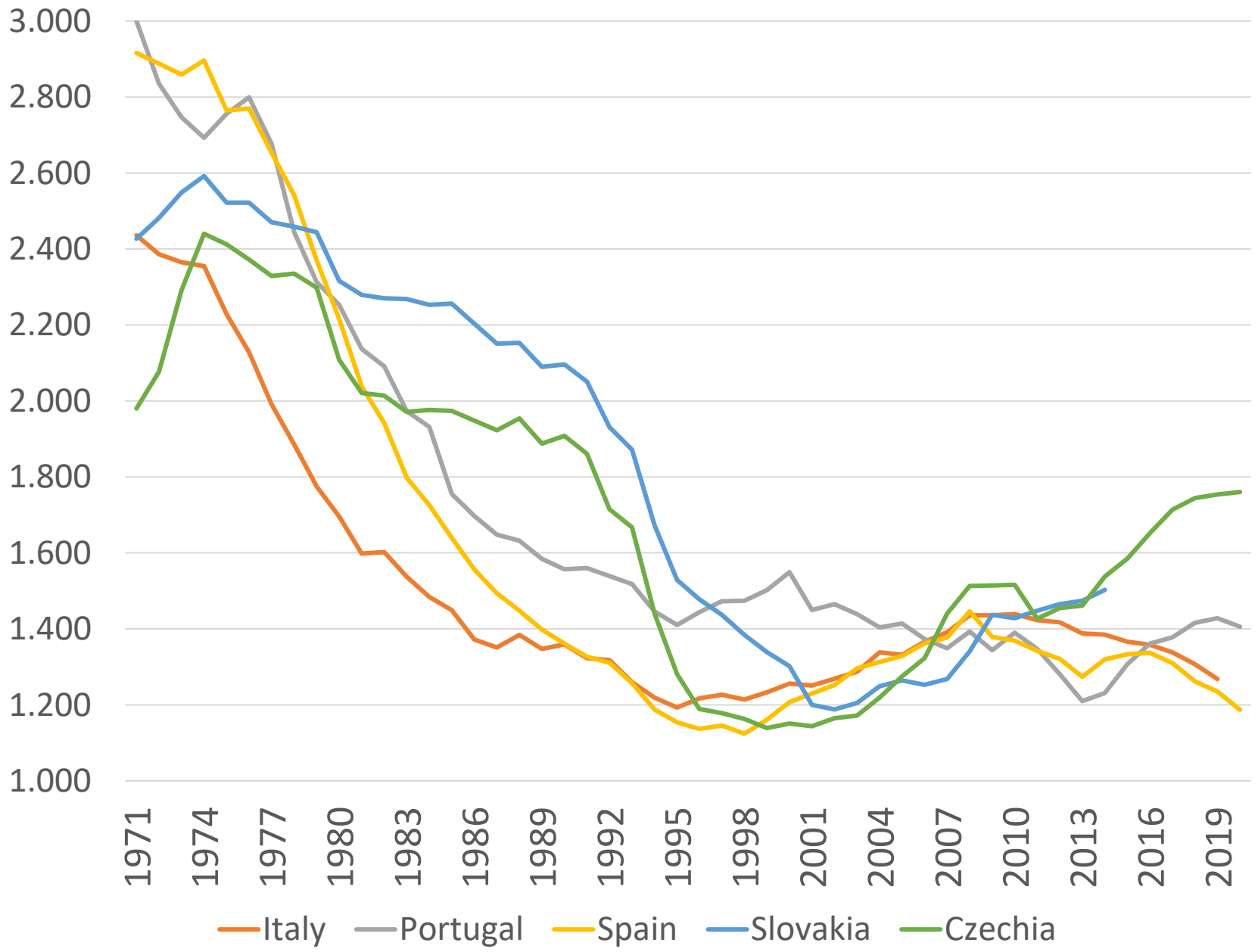
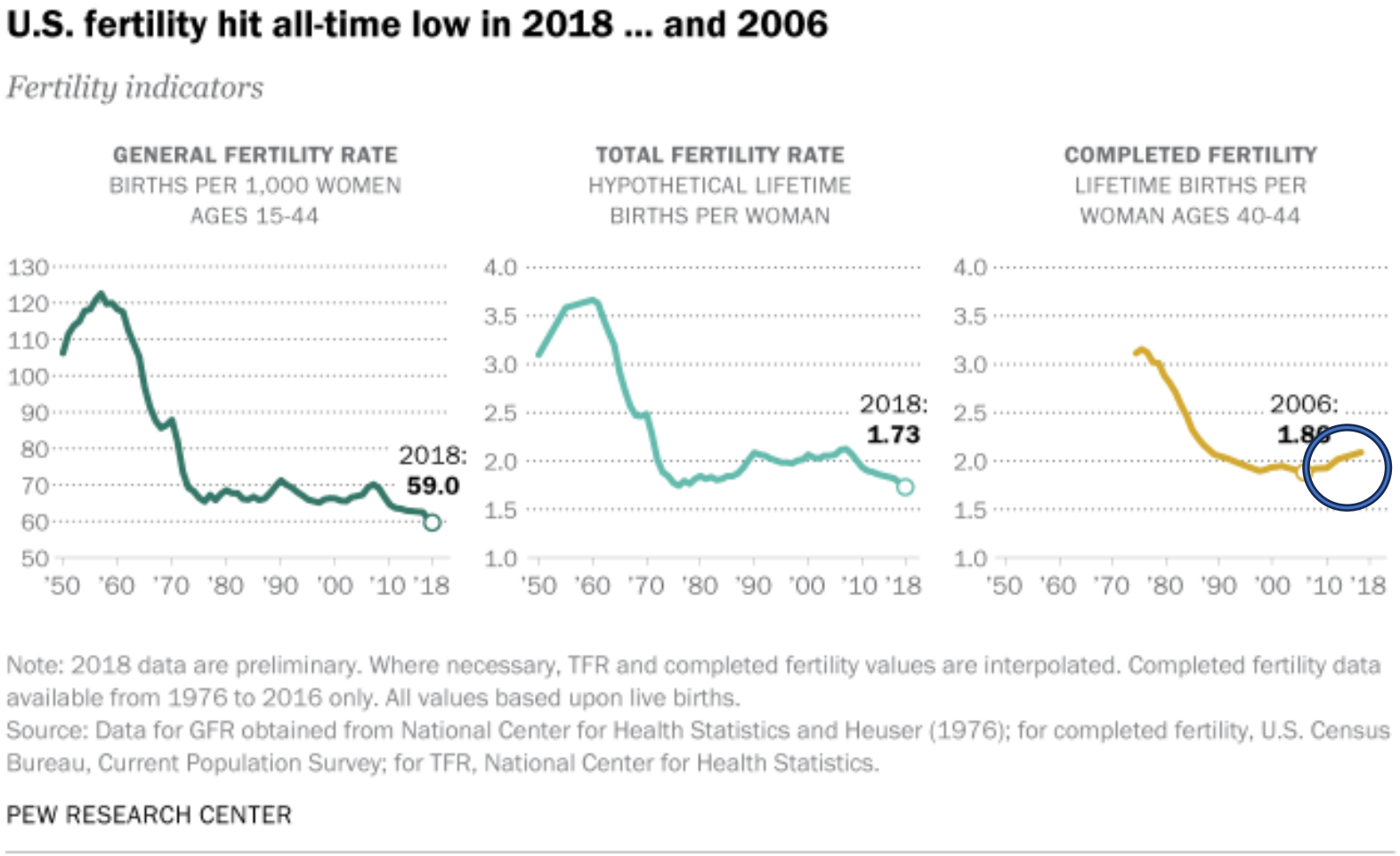


When cohort fertility keeps declining, TFR declines and stays low

Human Fertility Database 2023

So what does that tell us about US fertility?



Completed cohort fertility still increasing!

Planned number of children has held steady

Over half a century of surveys, female teenagers age 15-19 have been fairly consistent, reporting their intention to have an average of just over two children during their lifetime. Each point describes the stated intention of girls born within that five-year time period.

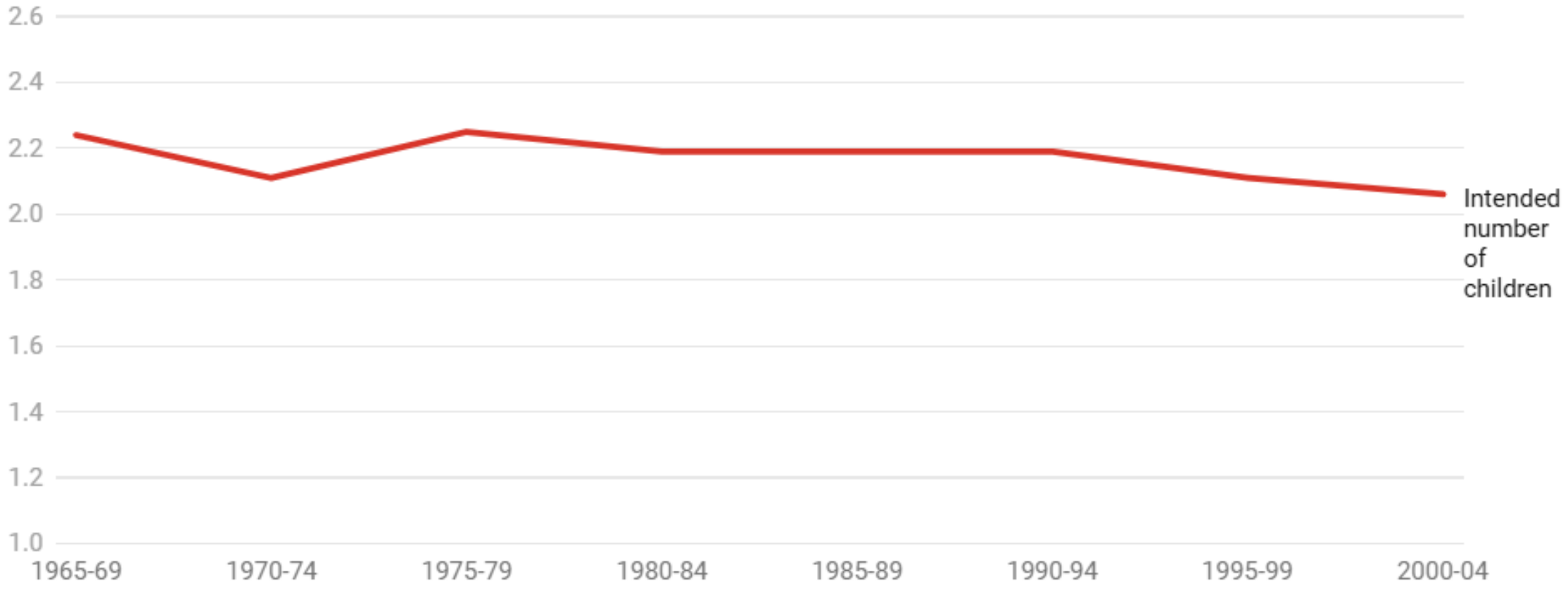
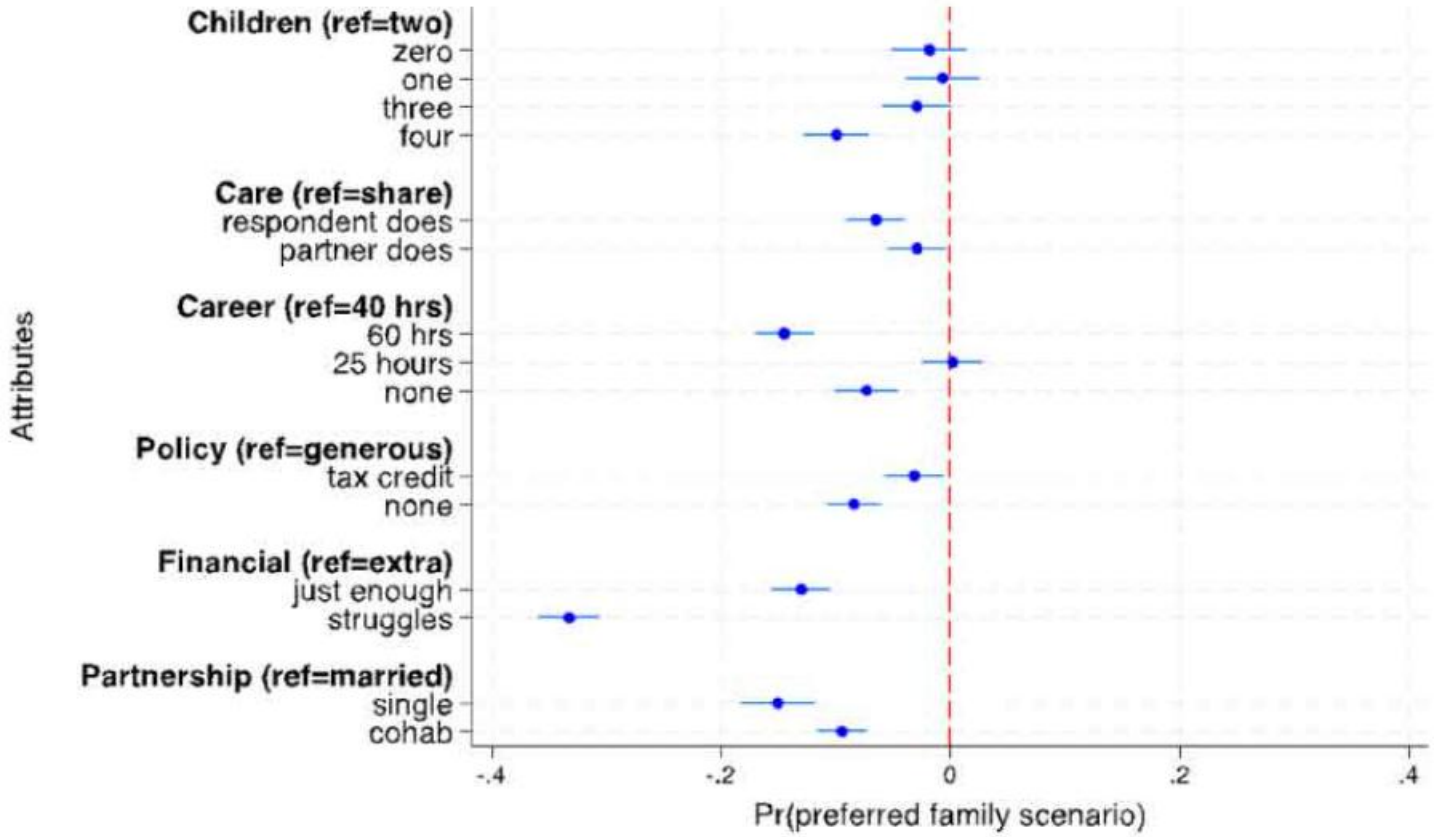


Chart: The Conversation, CC-BY-ND • Source: Guzzo and Hayford, "Evolving Fertility Goals and Behaviors in Current U.S. Childbearing Cohorts," Population and Development Review 2023 • Get the data • Download image • Created with Datawrapper

How strong are preferences for 2 children?

Figure 1: Effect of different family factors on the predicted probability of preferring alternative family scenarios (all other factors equal)



Source: Behrman, Marshall and Keusch (2025)

Preferences for 2 kids aren't significantly stronger than preferences for 0, 1, or 3 kids when work, financial, and family concerns are taken into account

Relationship status, work hours, financial security, and the availability of family-friendly policies had a stronger influence on preferred family scenarios than the number of children did

3. Age and Fecundity

- Big debate about the relationship between these two variables
- Can’t just look at modern fertility rates by age
  - Many older couples don’t want any more children
- Can’t just look at people who try fertility treatments either!
  - These people are *already* having trouble conceiving
- **Old data** are often most helpful!

Maximum fertility??

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By age 34, average of 5 pregnancies

Source: Tietze 1957

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Average of 7 pregnancies at ages 35+

Source: Tietze 1957

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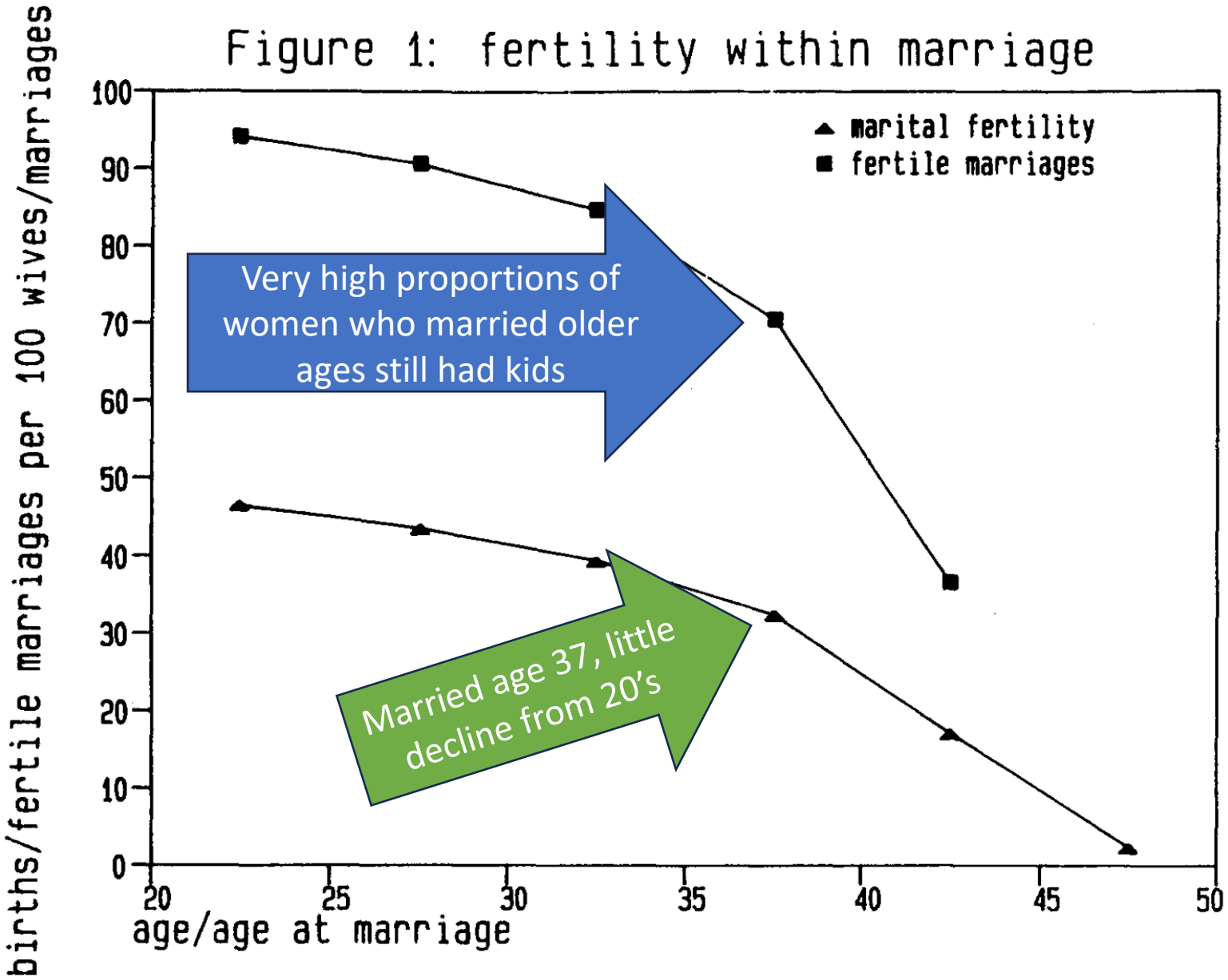
But average age at marriage was 22, so max is probably higher!

Source: Tietze 1957

17th Century Northwestern Europe:

Late marriage, but still high fertility



Fecundity with Age:
Often it just takes more trying!

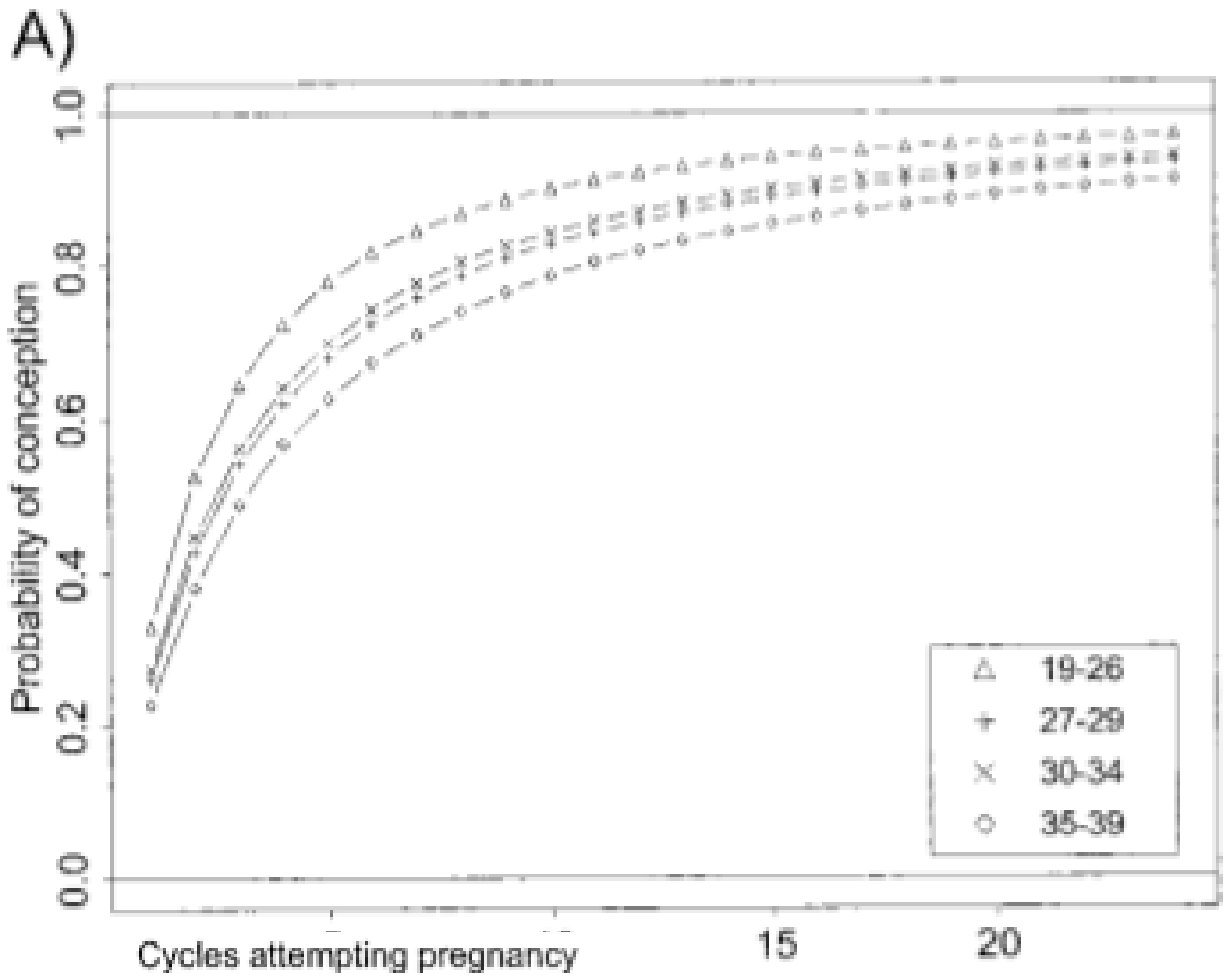
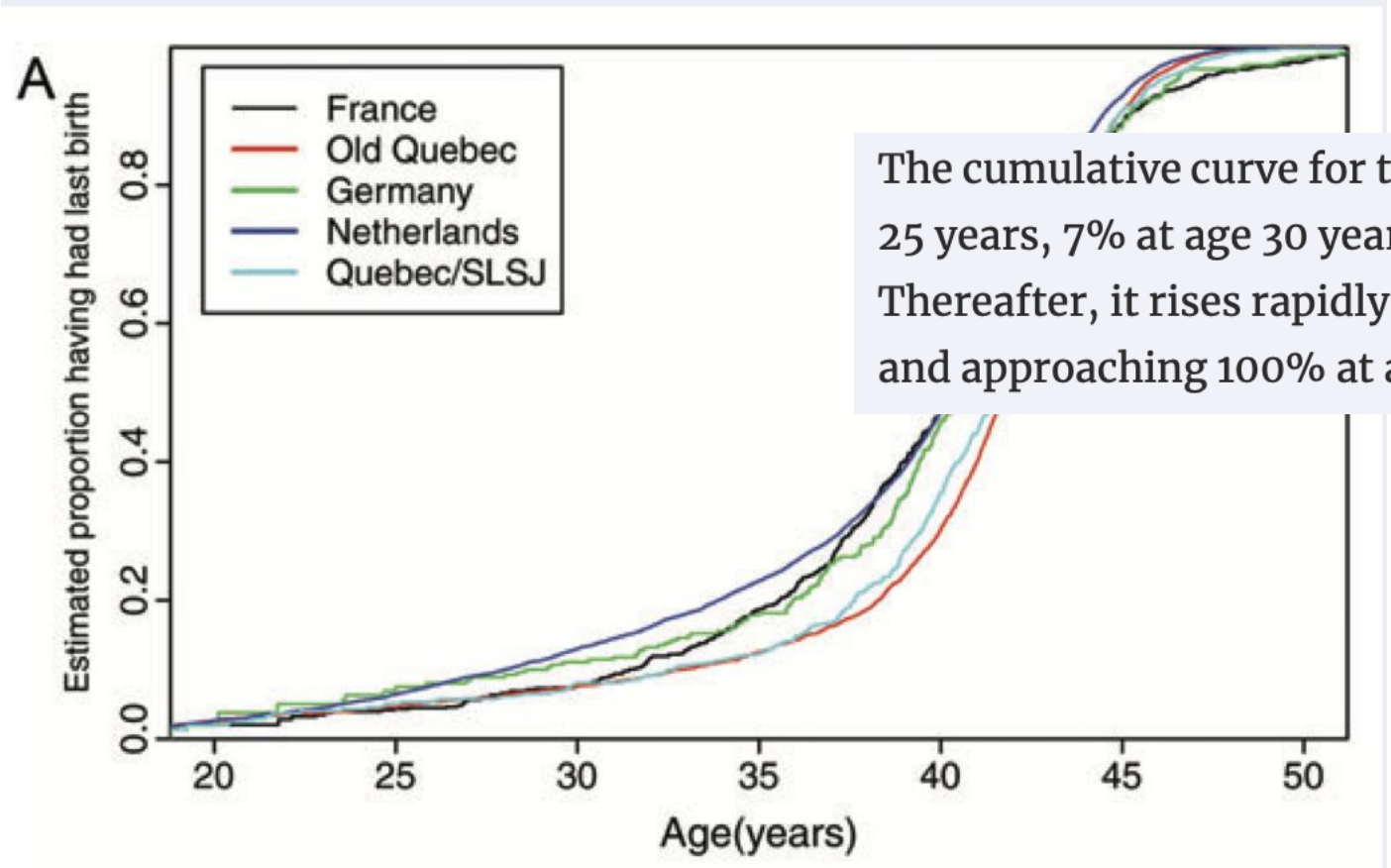


Figure 1. A) Cumulative probability of conceiving a clinical pregnancy by the number of menstrual cycles attempting to conceive for women in different age categories (assuming intercourse occurs at a frequency of 2 times week). B) Cumulative probability of conceiving a clinical pregnancy by the number of menstrual cycles attempting to conceive for women in different age categories (assuming intercourse occurs at a frequency of 1 time week).

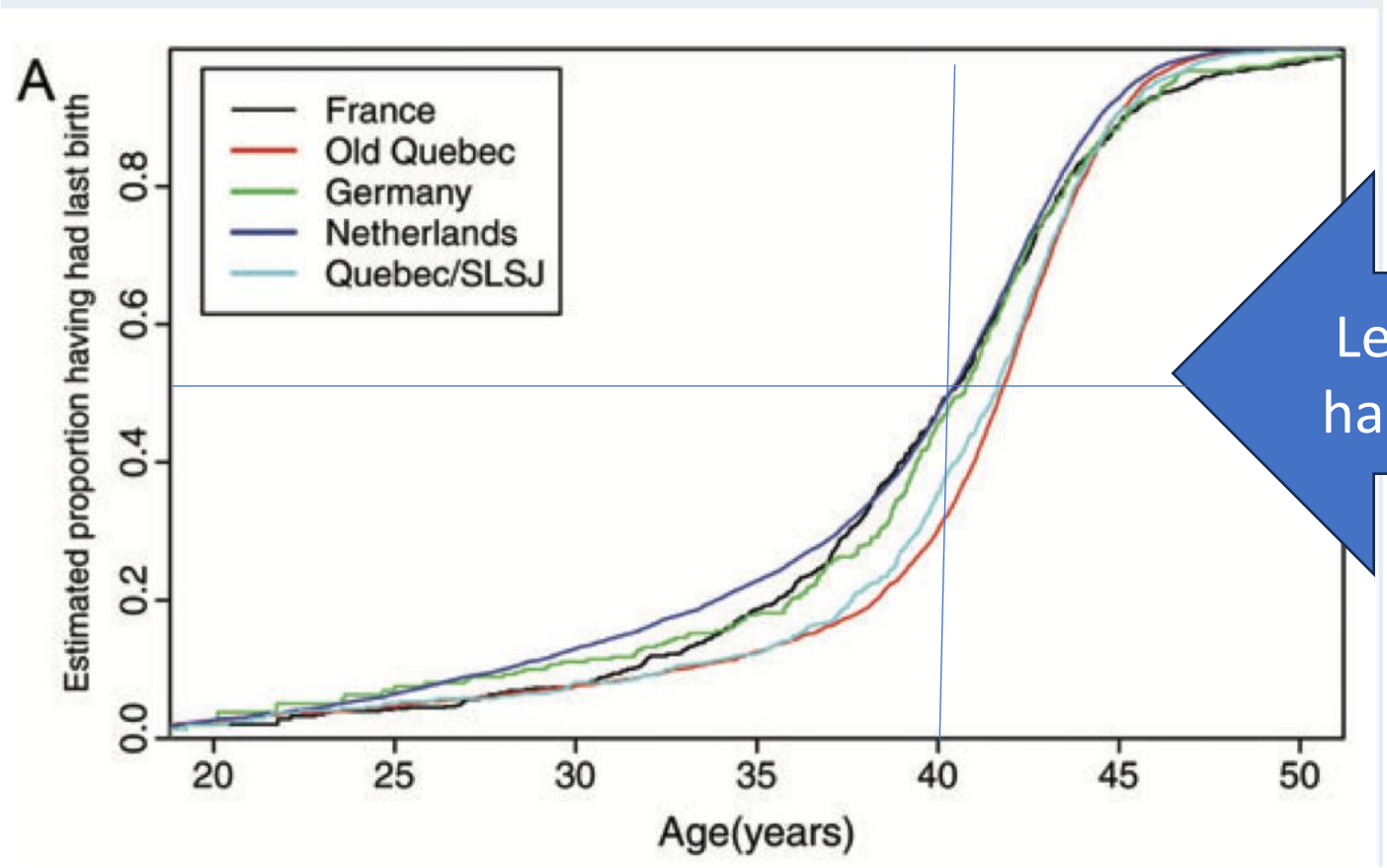
Dunson. Age and Infertility. Obstet Gynecol 2004.

From the Reading



The cumulative curve for the end of fertility slowly increases from 4.5% at age 25 years, 7% at age 30 years, 12% at age 35 years and 20% at age 38 years. Thereafter, it rises rapidly to about 50% at age 41, almost 90% at age 45 years and approaching 100% at age 50 years.

From the Reading

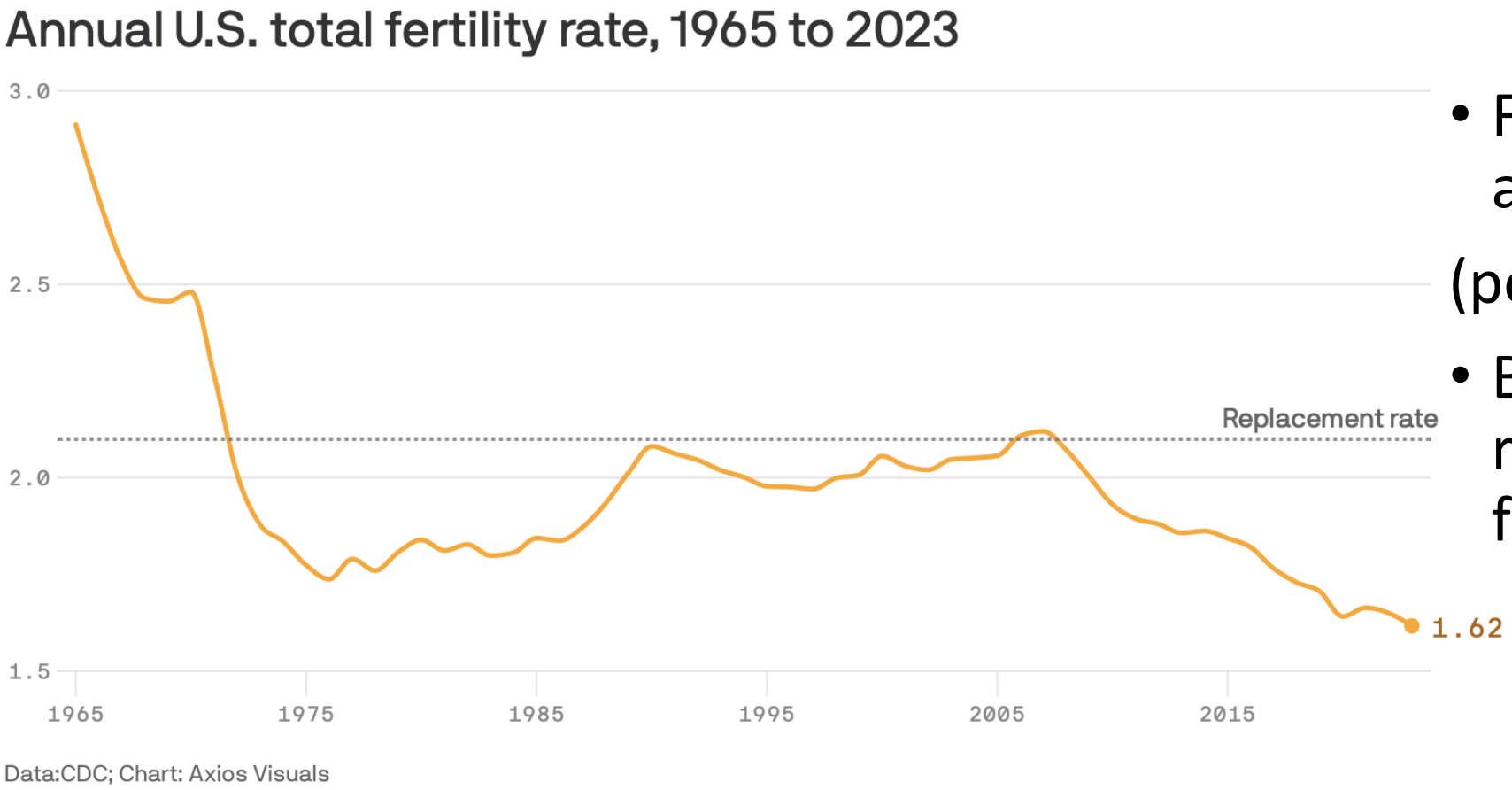


Less than 1/2 of people had had their *last* birth by age 40

3. Age and Fecundity

- Fecundity declines with age
  - Decline accelerates quickly in late 30’s and early 40’s
- But back in the day people routinely had babies into their 40’s

So what does this all mean for US fertility?



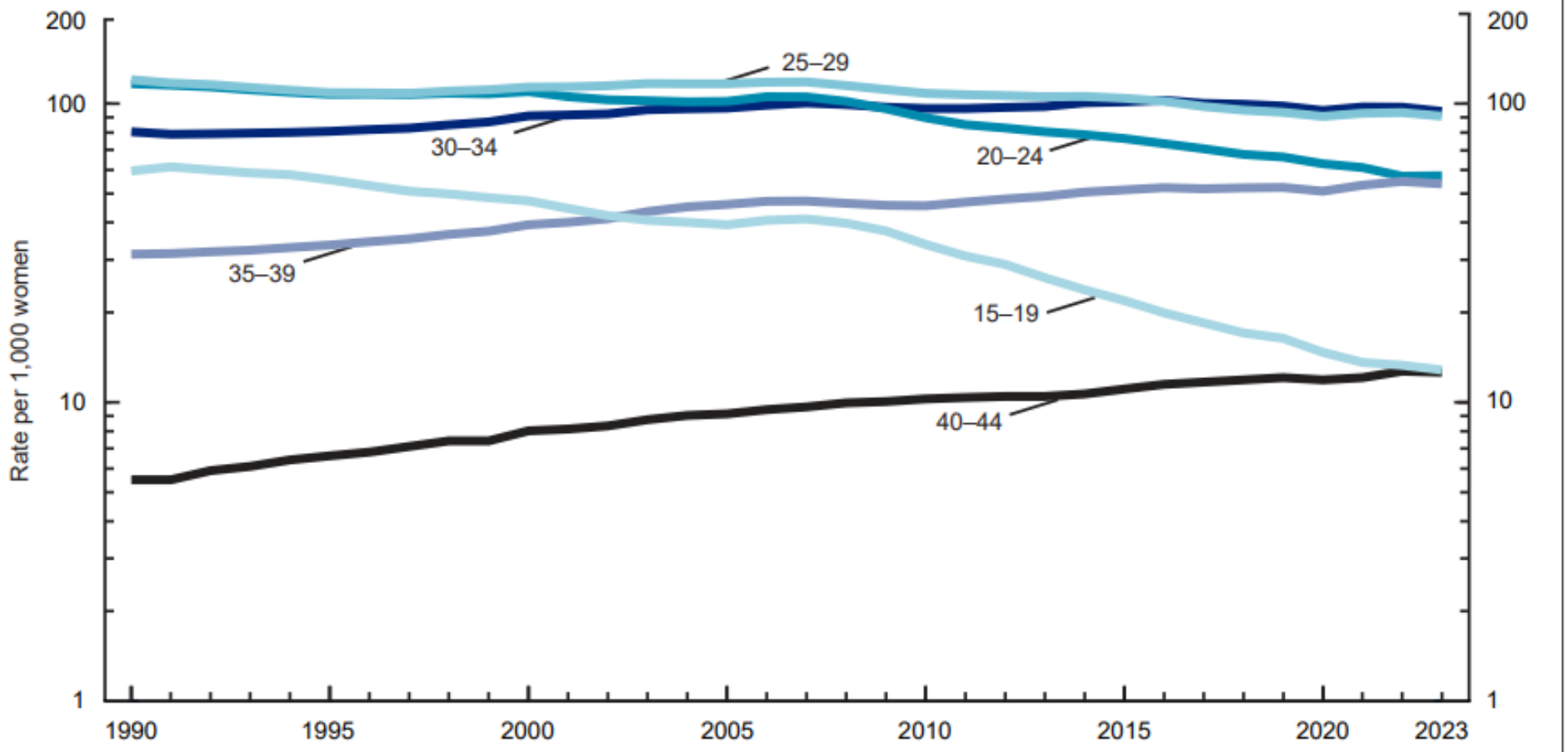
- Fertility rates can rebound after falling

(postponement transition)

- But the potential for rebound is limited by fecundity at later ages

Whose fertility is declining?

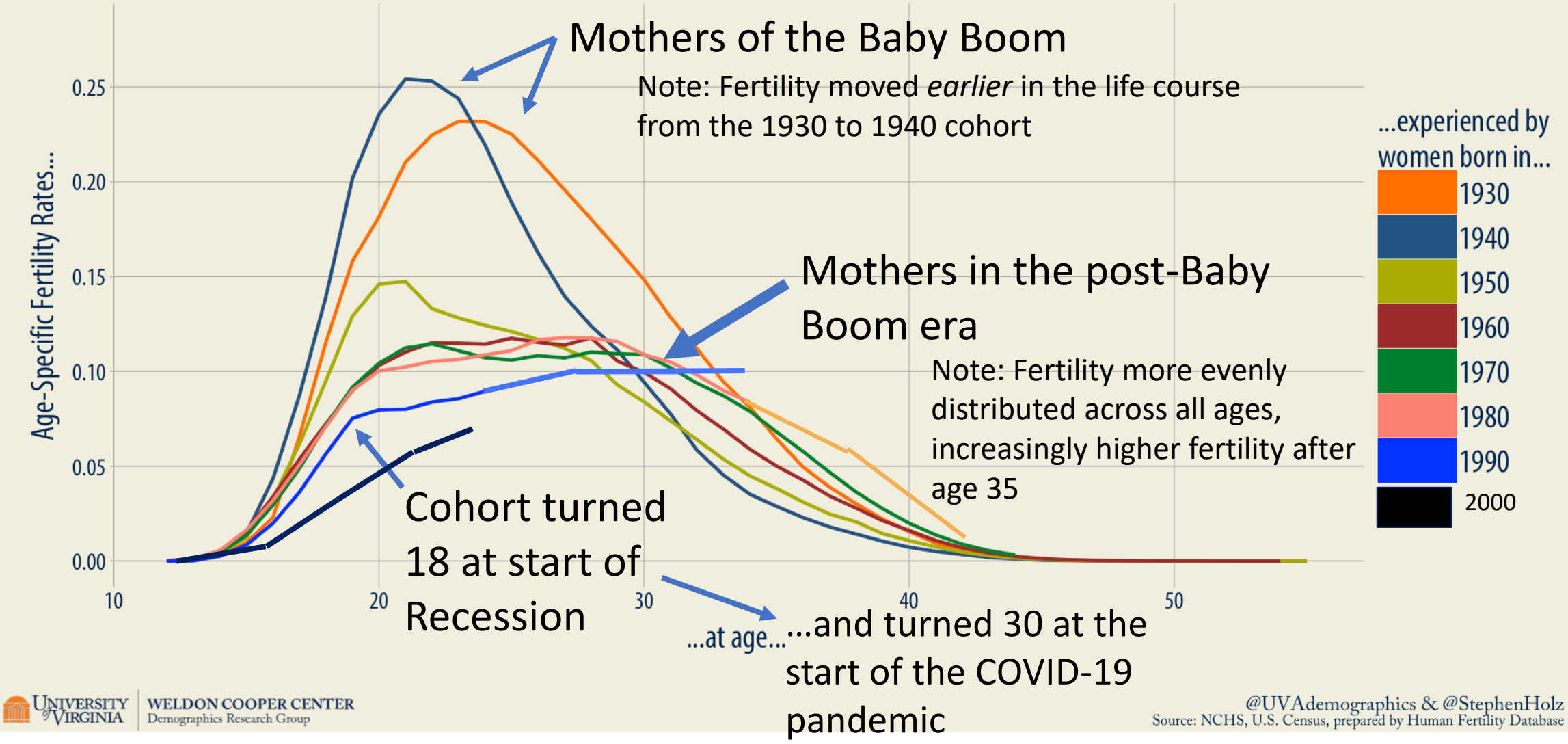
Figure 2. Birth rate, by selected age of mother: United States, 1990–2023



NOTES: Rates for 2020 have been revised and may differ from those published in "Births: Final Data for 2020." Rates are plotted on a logarithmic scale.
SOURCE: National Center for Health Statistics, National Vital Statistics System, natality data file.

Declining birth rate driven by declining **teen births** and births to **20-24 y.o. women**

USA ASFRs Experienced by Select Cohorts



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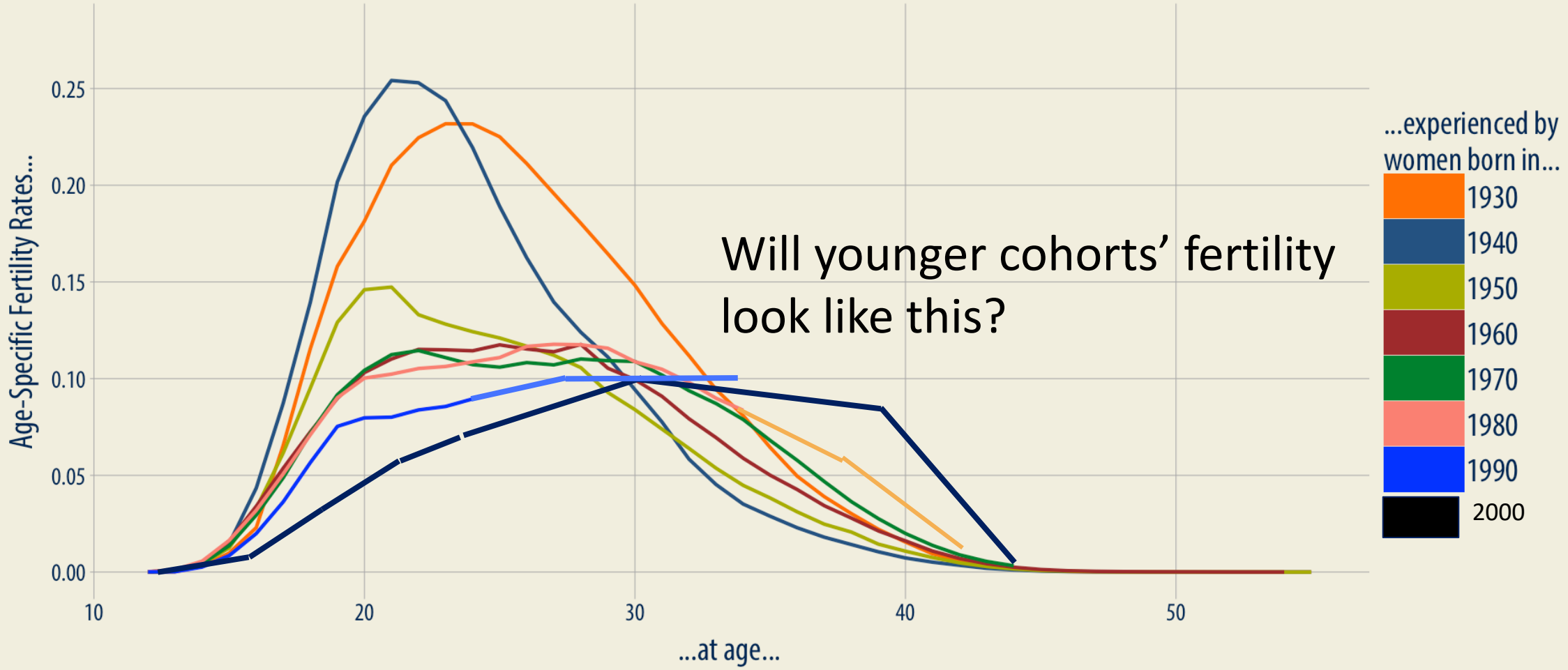
@UVAdemographics & @StephenHolz

Source: NCHS, U.S. Census, prepared by Human Fertility Database

USA ASFRs Experienced by Select Cohorts



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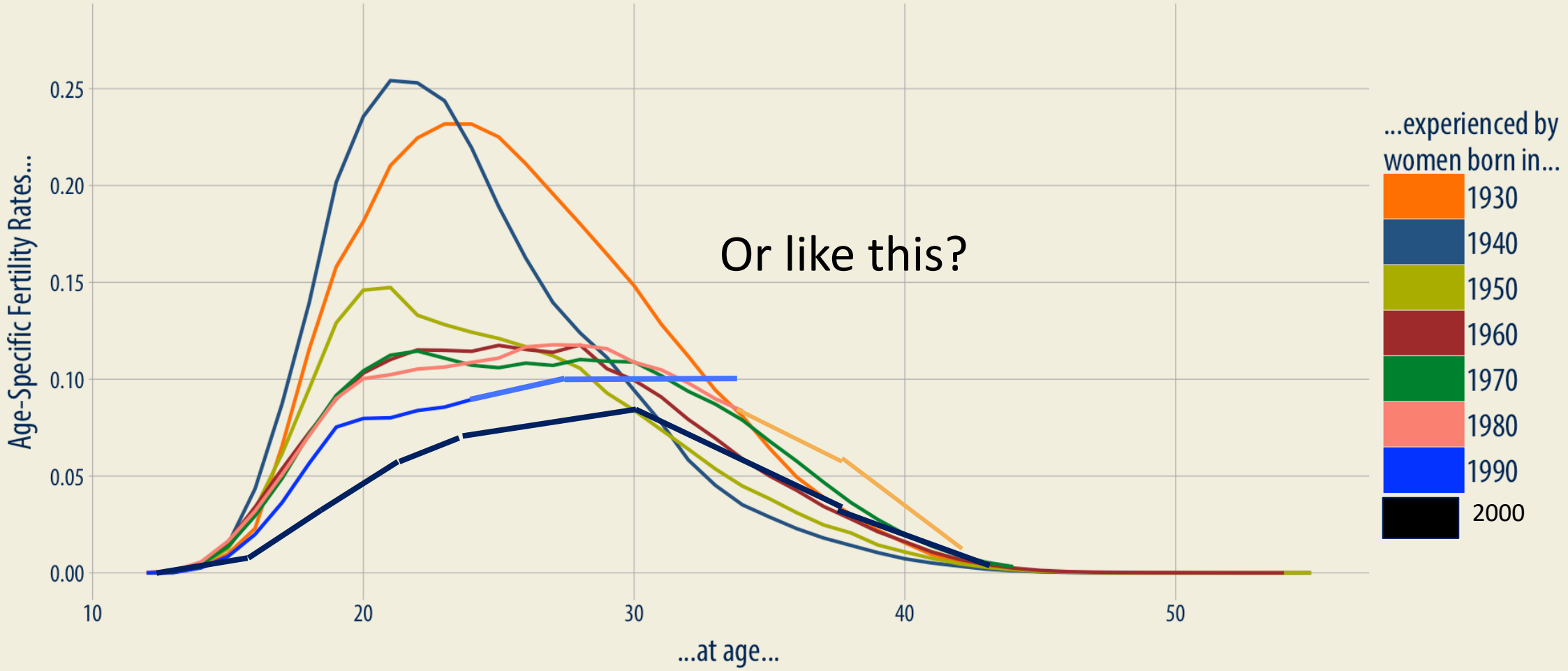
Source: NCHS, U.S. Census, prepared by Human Fertility Database

USA ASFRs Experienced by Select Cohorts

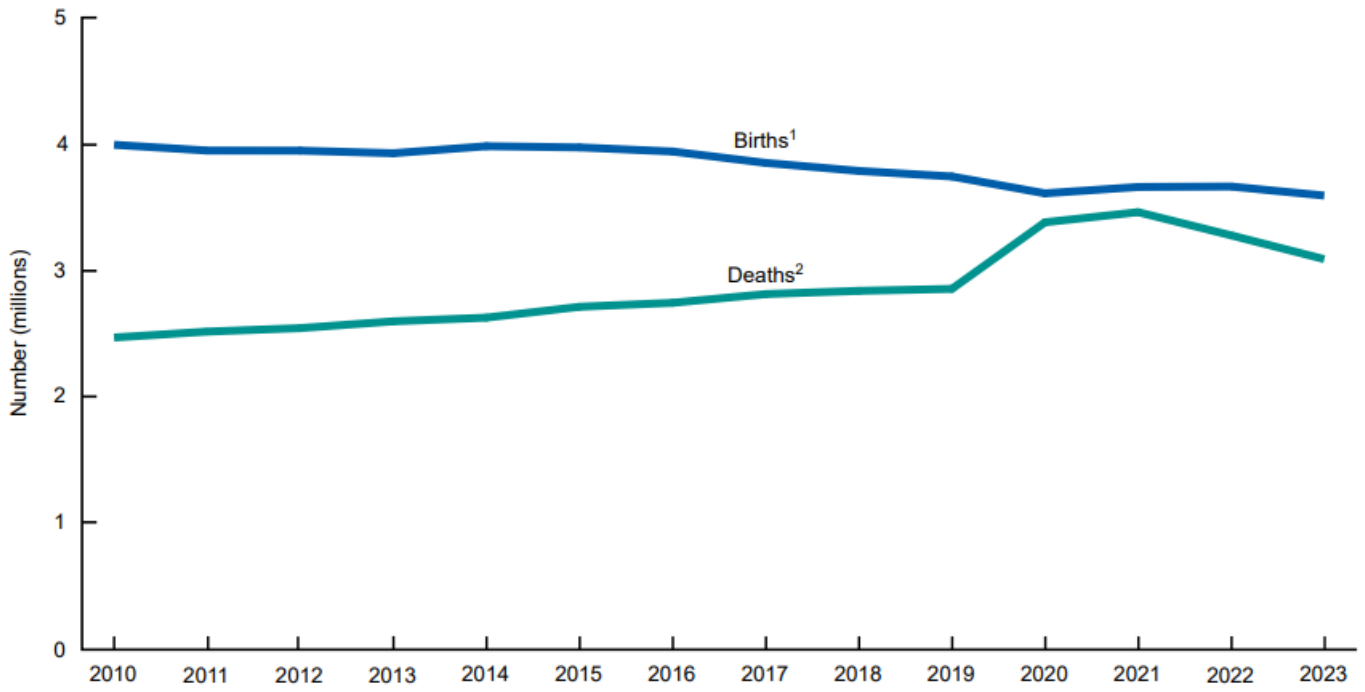


@UVAdemographics & @StephenHolz

Source: NCHS, U.S. Census, prepared by Human Fertility Database

Is the US heading towards population collapse?

Figure 1. Number of births and deaths: United States, 2010–2023



[1]No significant trend was observed from 2010 through 2016; a significant decreasing trend was observed from 2016 through 2019; no significant trend was observed from 2019 through 2023 ($p < 0.05$).
[2]A significant increasing trend was observed from 2010 through 2019; no significant trend was observed from 2019 through 2023 ($p < 0.05$).
SOURCE: National Center for Health Statistics, National Vital Statistics System, natality and mortality data files.

The United States is still in a period of natural increase where the number of births exceeds the number of deaths

(because of population momentum!)

Source: Hamilton, Driscoll, and Minino 2025

Question of the day

Which are more informative of fertility trends: period or cohort measures? Explain in 1-2 sentences

https://tinyurl.com/feb10soc170